

ENVIRONMENTAL MONITORING

CS4086

Environmental Data Science (EDS)

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ENVIRONMENTAL MONITORING

- Environmental monitoring is a tool to assess environmental conditions and trends, support policy development and its implementation, and develop information for reporting to national policymakers, international forums and the public.
- The monitoring of urban air pollution — an important human health risk — is poor in many cities of the subregion. Solid and hazardous waste monitoring is weak and industrial emissions are also not well monitored, reducing the effectiveness of policy instruments such as emissions charges and fines. Monitoring of transboundary air pollution also needs strengthening.

FIVE IMPORTANT GLOBAL-SCALE ENVIRONMENTAL INDICATORS

- biological diversity,
- food production,
- average global surface temperature and atmospheric CO₂,
- human population, and
- resource depletion

FIFTH ENVIRONMENT FOR EUROPE CONFERENCE (2003)

- UNECE (United Nations Economic Commission for Europe) Ministers endorsed **recommendations for strengthening environmental monitoring** and information systems in European and Central Asian countries

prepared by the UNECE Working Group on Environmental Monitoring. It is available:

https://unece.org/sites/default/files/2020-12/Final.English.Rev_.2.01.12.05.pdf

WHY IS ENVIRONMENTAL MONITORING SO IMPORTANT?

Protecting ecosystems

Environmental monitoring can help us understand the **health of ecosystems** and changes in **biodiversity**.

By monitoring environmental indicators, we can detect ecological problems in time and take corresponding protective measures to ensure the stability and diversity of ecosystems.

Maintaining human health

Environmental monitoring can detect **harmful substances and pollutants in** the environment such as air, water and soil.

Timely detection and treatment of pollution problems can prevent and reduce the impact of environmental pollution on human health.

WHY IS ENVIRONMENTAL MONITORING SO IMPORTANT?

Safeguarding drinking water

By monitoring water quality and the **condition of water sources**, we can ensure that drinking water is safe and hygienic, and prevent water sources from being polluted or overexploited.

Responding to climate change

Environmental monitoring can track **climate indicators and environmental changes**, helping scientists and governments to understand the trends and impacts of climate change, and to formulate **policies and measures to address climate change**.

WHY IS ENVIRONMENTAL MONITORING SO IMPORTANT?

Supporting environmental management decisions

Environmental monitoring provides scientific environmental data and information to support **environmental management decisions**.

The government, enterprises and the public can formulate appropriate **environmental protection** and improvement measures based on the monitoring data.

Promoting sustainable development

By monitoring environmental quality and **natural resource utilization**, we can achieve rational utilization of resources and sustainable development, **avoiding over-exploitation** and **environmental damage**.

WHY IS ENVIRONMENTAL MONITORING SO IMPORTANT?

Early warning of natural disasters

Environmental monitoring can be used to monitor the **precursors of natural disasters**, such as **earthquakes and floods**, to provide early warning information and reduce disaster losses.

Global cooperation and data sharing

Environmental monitoring involves environmental issues on a global scale, and there is a need for cooperation and data sharing among countries to address global environmental challenges.

APPLICATION AREAS FOR ENVIRONMENTAL MONITORING

- There are many different types of environmental monitoring, each designed to measure specific environmental parameters. **Some common types of environmental monitoring include:**
- **Water resources management:** monitoring **water quality, water quantity and water level**, guaranteeing rational utilization and management of water resources.
- **Atmospheric environment monitoring:** detecting air quality, warning air pollution and improving **urban air quality**.
- **Soil monitoring:** monitoring soil pollution, protecting **farmland** and **ecological** environment.
- **Noise and radioactivity monitoring.**

ENVIRONMENTAL MONITORING OF WATER

- **Physical parameters:** these include temperature, pH, turbidity, conductivity and color.
- **Chemical parameters:** these include dissolved oxygen, nutrients, metals and organic contaminants.
- **Biological parameters:** these include bacteria, algae and other microorganisms.

PH

pH is an indicator of the acidity or alkalinity of a body of water, with values ranging from 0 to 14.

A pH value of **7 means that the water is neutral**,
a pH value of **less than 7 means that it is acidic**, and
a pH value of **more than 7 means that it is alkaline**.



A10 Aquarium ORP pH Controller can monitor

TEMPERATURE

Temperature is an important parameter because it affects the solubility of gases in water.

For example, dissolved oxygen levels decrease as temperature increases.

It is measured using thermometers or temperature sensors, commonly glass thermometers, digital thermometers, and infrared thermometers.

CONDUCTIVITY

Conductivity is a measure of the ability of water to conduct electricity.

It is caused by ions dissolved in the water, such as sodium, chloride and calcium.

Measured with a **conductivity meter**, they measure the ability of water to conduct electricity, which gives the concentration of ions in the water.



A20 EC Water Conductivity Tester

COLOR

Color is a subjective measure of the appearance of water.

It can be caused by dissolved organic matter or minerals (e.g., iron).

It is measured using a colorimeter or spectrophotometer, which determines the shade of color of a water sample by comparing its color to a standard color.



DISSOLVED OXYGEN

Dissolved oxygen is the **amount of oxygen dissolved in water.**

It is critical to the survival of fish and other aquatic organisms.

DO low levels can impact our health

It is measured using a Dissolved Oxygen Meter or Dissolved Oxygen Sensor which measure the amount of dissolved oxygen in the water.



NUTRIENTS

Nutrients are substances that are essential for **plant growth**.

They include **nitrogen, phosphorus and potassium**.

Excessive levels of nutrients can lead to **algal blooms**, which can reduce water quality.

They are measured using chemical analysis instruments, for **example nitrogen and phosphorus can be measured by analyzing the chemical composition of a water sample**.

A harmful algal bloom occurs when toxin-producing algae grow excessively in a body of water.

The excessive algal growth, or algal bloom, becomes visible and can be green, blue-green, red, or brown, depending on the type of algae.



METALS

Metals are naturally occurring elements found in water.

Some metals, such as **lead and mercury**, are toxic to humans and other animals.

Measured using an **atomic absorption spectrometer (AAS)** or an **inductively coupled plasma emission spectrometer**, they can accurately measure the amount of metallic elements in water.

AAS is an analytical technique used to determine the concentration of metal atoms/ions in a sample.

Inductively Coupled Plasma Optical Emission spectroscopy (ICP-OES) is an analytical technique used to determine how much of certain elements are in a sample.



ORGANIC POLLUTANTS

- Organic pollutants are compounds made up of **carbon**.
- They can be released into water from a variety of sources, such as **industrial wastewater, agricultural runoff, and sewage**.
- Measured using gas chromatography mass spectrometry or high performance liquid chromatography, these instruments are capable of analyzing organic substances in water samples.
- By monitoring the water level and flow of rivers, lakes and reservoirs, the availability of water resources can be **grasped in real time**, providing data support for water resource management and scheduling.

Gas chromatography/mass spectrometry (GC/MS) combines two analytical tools to identify and measure the concentration of chemicals found in foods, consumer products, pharmaceuticals, fuels, the environment, and more.

HPLC separates compounds dissolved in a liquid sample and allows qualitative and quantitative analysis of what components and how much of each component are contained in the sample.



WATER FLOW METER

A water flow meter is a device used to measure water flow, and common water flow meters include **vortex flow meters**, **electromagnetic flow meters**, and **ultrasonic flow meters**.

They calculate the water flow rate by measuring the **flow rate and cross-sectional area of the pipe section** through which the water flows.



WATER LEVEL METER

Water level meters are used to measure the height of the water level of a body of water, so as to deduce the volume of the body of water and the amount of water.

Common types of water level gauges are pressure sensors, float type water level gauges, etc.

Float level gauges are simple devices that measure liquid content in the likes of ISO containers, domestic tanks, bottles, storage tanks, trucks and trailers. A float level gauge is also greatly used in compressors, generators and agricultural and construction machinery.

HYDROLOGICAL METHODS

Hydrological methods calculate the volume of water in a water body by collecting hydrological data such as:

- rainfall,
- evaporation, and
- seepage.

FLOW STATIONS

A flow station is a type of observatory set up in water bodies such as rivers and lakes to monitor data such as water flow and water level.

ENVIRONMENTAL MONITORING OF AIR

- **Particulate Matter (PM)** is a mixture of solid particles and liquid droplets suspended in the air. It can be hazardous to human health and may also contribute to climate change.
- **Ozone (O₃)** is a gas formed when pollutants react in sunlight. It can be hazardous to human health and may damage plants and materials.
- **Sulfur Dioxide (SO₂)** is a gas formed when sulfur-containing fuels are burned. It can be hazardous to human health and can cause acid rain.
- **Nitrogen Dioxide (NO₂)** is a gas formed when nitrogenous fuels are burned. It can be harmful to human health and can cause smog.
- **Volatile Organic Compounds (VOC):** compounds that evaporate easily. They can be harmful to human health and may contribute to smog.
- **Carbon Monoxide (CO)** produced when fuels are burned incompletely. It is harmful to human health and may cause traffic congestion.
- **Radon Gas** is a radioactive gas that exists underground. It may be hazardous to human health and may cause lung cancer.

ROBUST SO2 AIR QUALITY MONITOR

- Sulfur dioxide has been measured in air, in water, and in food and beverages.
- Methods for determining levels in the air include:
 - ❖ ion chromatography,
 - ❖ titration,
 - ❖ calorimetry,
 - ❖ mass spectrometry,
 - ❖ conductimetry,
 - ❖ amperometric detection,
 - ❖ flame photometric detection, and
 - ❖ turbidimetry.
- This SO2 gas detector uses **Electrochemical Sensing** technology to measure the total SO2 in the air.
- The monitor is equipped with an **SO2 sensor module**.



INDUSTRIAL OZONE METER

- The Model 106-H is a high concentration ozone analyzer designed to control the output of an ozone generator.
- This instrument is designed to be installed in-line with an ozone generator and ozone process.



<https://2btech.io/items/industrial-ozone-monitors/model-106-h-ozone-monitor/>

PM10 / PM2.5 PORTABLE PARTICULATE MONITOR

The Aeroqual PM10 / PM2.5 Portable Particulate Monitor provides accurate and simultaneous real-time measurement of PM₁₀ and PM_{2.5} in ambient outdoor and indoor environments.

Measure gaseous pollutants by simply swapping the PM sensor head for the gas sensor head of your choice (gas sensor heads available separately).

Suitable for use on a range of projects, **including indoor / outdoor air quality testing, personal exposure monitoring, and short-term fixed monitoring.**



HOW DO CARBON MONOXIDE DETECTORS WORK?

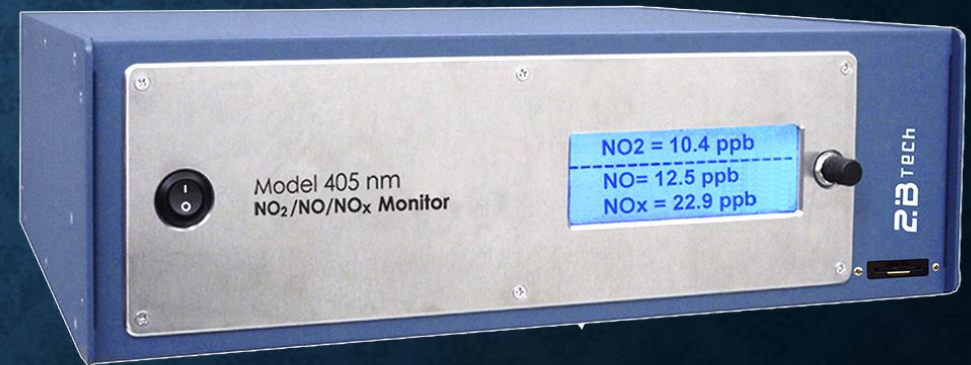
- Carbon monoxide detectors sound an **alarm** when they sense a certain amount of carbon monoxide over time. Different sensors set off different types of alerts:
- **Biomimetic sensor:** a gel changes color when it absorbs carbon monoxide, and this color change triggers the alarm.
- **Metal oxide semiconductor:** when the silica chip's circuitry detects carbon monoxide, it lowers the electrical resistance, and this change triggers the alarm.
- **Electrochemical sensor:** electrodes in a chemical solution sense changes in electrical currents when they come into contact with carbon monoxide, and this change triggers the alarm.
- Once the carbon monoxide detector alarm sounds, it must be in a carbon monoxide-free environment to silence the siren.

NO₂/NO/NO_x MONITOR

An approved NO₂ monitor that uses the technique of direct absorption of 405 nm light by NO₂ molecules.

The Model 405 nm also measures NO by conversion to NO₂ using ozone.

The instrument provides a total measurement of NO_x as the sum of NO and NO₂.



ENVIRONMENTAL MONITORING OF SOILS

- It varies depending on the specific application. However, some **common parameters** include:
- **Physical parameters:** these include texture, structure, pH and moisture content.
- **Chemical parameters:** these include organic matter content, nutrient content and metal content.
- **Biological parameters:** these include microbial activity, earthworm abundance and plant diversity.

FREQUENTLY MONITORED PARAMETERS IN SOIL MONITORING:

- **Texture** refers to the **size and shape** of soil particles. It is an important parameter as it affects the **drainage, aeration and water holding capacity of the soil**.
- **Structure** is the **arrangement** of soil particles, which affects **soil stability and the movement of water and air through the soil**.
- **Water content: Moisture** content is the amount of water present in the soil, which affects **plant growth** and the movement of nutrients through the soil.
- **Organic matter content** is the amount of organic matter present in the soil. It is an important parameter as it affects **the fertility of the soil and the ability of the soil to retain water**.
- **Nutrient content** are substances essential for plant growth. They include nitrogen, phosphorus and potassium. Nutrient content is an important parameter as it affects the productivity of the soil.
- **Metal content** are naturally occurring elements in the soil. Some metals, such as lead and mercury, are toxic to humans and other animals. Metal content is an important parameter because it affects the safety of soil for human use.
- **Biological parameters** such as **microbial activity and earthworm abundance** are important indicators of soil health. They can also be used to assess the **potential for human health risks**.

ENVIRONMENTAL MONITORING OF NOISE

- It varies depending on the specific application. However, some common parameters include:
- **Sound pressure level (SPL)**
SPL is a measure of the loudness of a sound and is measured in decibels (dB).
- It is measured in **decibels (dB)**; the lower the SPL, the quieter the sound; the higher the SPL, the louder the sound.

NOISE MONITORING

- The spectrum of a sound is a **measure of the distribution of energy at different frequencies**. It is measured in Hertz (Hz).
- The lower the frequency, the lower the pitch of the sound. The higher the frequency, the higher the pitch of the sound.
- **Time-varying characteristics**
The time-varying characteristic of sound is a measure of how the sound level changes over time.
- This is important for assessing the effects of noise on **human health**, as some types of noise can be more harmful than others.
- **Source location**
The location of the noise source is important for identifying the source and taking steps to reduce it.



ENVIRONMENTAL MONITORING FOR RADIOACTIVITY

- Radiological monitoring is the process of measuring and recording levels of **radioactivity in the environment**.
- It is used to assess the potential for exposure to radiation and to enforce radiation safety regulations.



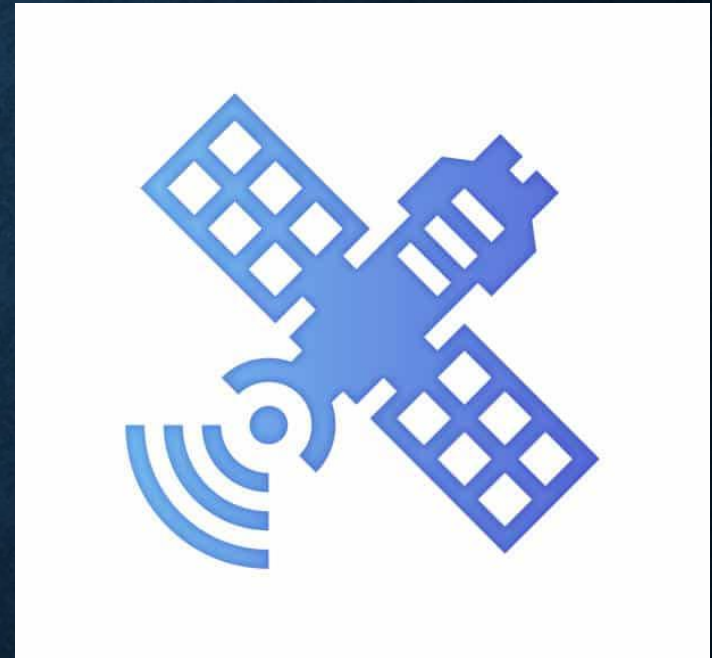
REMOTE SENSING

- **Air monitoring**, water monitoring and soil monitoring can be done by manual sampling, continuous monitoring and remote sensing.
- **Manual sampling** is the most common method of environmental monitoring. It involves the collection of air, water or soil samples which are then analyzed in a **laboratory**.
- **Continuous monitoring** involves the use of sensors to continuously measure the concentration of pollutants in air, water or soil.
- Remote sensing involves the use of **satellites or airplanes** to measure the concentration of **pollutants in air, water or soil**.

REMOTE SENSING

In environmental monitoring, remote sensing is used to measure the concentration of pollutants in air, water or soil. This is done by **using sensors** to measure the amount of radiation reflected or emitted by the object or area being monitored. The type of sensor used depends on the particular pollutant being measured.

Remote sensing can be used to track air pollution levels and identify areas with high levels of pollution, to track water quality levels and identify areas that are suffering from water pollution, and to understand the **impact of human activities on the environment and to plan future land use.**



RADIATIONS

- **Alpha radiation** is the type of radiation emitted by certain radioactive elements. It is the **least penetrating** type of radiation, but may be harmful if ingested or inhaled. Source: soil
- **Beta radiation** is a type of radiation emitted by certain radioactive elements. It is **more penetrating than alpha radiation**, but still not very penetrating. It is usually found in the air, especially near nuclear power plants or other radioactive sources. Water. Soil. Food.
- **Gamma radiation** is a type of radiation emitted by certain radioactive elements. It is **the most penetrating type of radiation** and may cause injury if exposed to the skin or internal organs. It is usually found **near nuclear power plants** or other radioactive sources. Water. Soil. Food.
- **Radon** is a **naturally occurring radioactive gas**. It is the second leading cause of lung cancer after smoking. It is found in the air, especially in areas with high concentrations of uranium in the soil. It is also found in water, especially groundwater. Soil. In food, especially food grown in areas with high concentrations of uranium in the soil.

DATA MANAGEMENT

Data management It is recommended that a national water information system, as a subsystem of the national water management system, should be updated or established to implement the following main tasks:

- (a) Collection of data on water quality and quantity (e.g., core network, and specialized networks);
- (b) Processing of the data (quality control);
- (c) Modelling of concentration fields of pollutants and of hydrological conditions;
- (d) Assessment and modelling of trends in water quality and quantity;
- (e) Assessment of health and environmental effects;
- (f) Emergency and warning in case of accidental pollution, possible health threat, and in case of extraordinary weather events accompanied by floods or droughts;
- (g) Reporting (both national and international);
- (h) Providing information to the public.

DATA MANAGEMENT

It is recommended that a national air quality information system, as a subsystem of the national air quality assessment and management system, should be established/updated to implement the following main tasks:

- (a) Collection of data on air quality (e.g., core network, specialized networks and mobile stations);
- (b) Processing of the data (quality control);
- (c) Modelling of concentration fields of pollutants;
- (d) Assessment and modelling of trends in air quality;
- (e) Assessment of health and environmental effects;
- (f) Reporting (both national and international);
- (g) Providing information to the public.
- (k) Preparation of emission projections.

REPORTING

Such interpretation should cover at least the following issues:

- (a) Populations living in areas with increased concentrations of pollutants in surface water and groundwater;
- (b) Areas of environmental importance with increased water-pollution levels;
- (c) Potential risks for human health and for the environment;
- (d) The origin of water pollution (both sectoral and territorial distribution of point and diffuse sources of pollution);
- (e) The impact of hydrological and meteorological conditions;
- (f) Trends in water quantity;
- (g) Trends in water pollution;
- (h) Policies applied and measures taken or proposed.

REPORTING

- (a) Population living in areas with increased concentrations of pollutants;
- (b) Areas of environmental importance (e.g., national parks or other protected areas) with increased pollution levels;
- (c) Potential risks for human health and for the environment;
- (d) Origin of air pollution (both sectoral and territorial distribution of emission sources);
- (e) Impact of meteorological conditions

SENSORS

- **Humidity sensors.** When the environment is too dry, static electricity can build up. Too humid, and equipment can begin to corrode. Since extreme humidity levels either way can damage equipment and cause downtime and humidity can change depending on outdoor conditions, we must monitor for humidity and maintain a stable environment
- **Airflow sensors.** Proper airflow in a data center will help avoid hot spots and maintain a stable ambient temperature, but cabling and other obstructions can build up over time and disrupt airflow. Monitor airflow to ensure cold air is efficiently cooling the environment or to understand if we need to make adjustments.
- **Pressure sensors.** the difference in airflow between the aisles can potentially cause certain partitions like plastic curtains to be draw into an aisle creating an air leak that can lead to inefficient cooling and hot spots. Similarly, a chimney above a high-heat, high air pressure rack may leak hot air into a neighboring chimney above a low-heat, low air pressure rack. Differential air pressure sensors are typically placed at the top and bottom of racks, between aisles, between underfloor perforated titles, and in vents and air plenums.

SENSORS

- **Water leak detection sensors.** Early detection of water in the data center can give you enough time to prevent a potential disaster. Consider deploying water sensors to avoid downtime caused by undetected air conditioning leakage, condensation, burst pipes, or local plumbing failures.
- **Contact closure sensors.** Keep your data center secure with contact closures to know if your cabinet doors are open or closed. Often, contact closures are connected to third-party sensors, such as a smoke detection sensor or a webcam that takes a picture when a cabinet door is opened.

REFERENCE

Guidelines for Developing National Strategies to Use Air and Water Quality Monitoring as Environmental Policy Tools UNITED NATIONS, New York and Geneva, 2012